



Catholic Junior College
JC 2 Preliminary Examinations
Higher 1

CHEMISTRY

Paper 1 Multiple Choice

8873/01

29 August 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, HT group and NRIC/FIN number on the Answer Sheet in the spaces provided.

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

For each question there are **four** possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 Rubidium is a soft, metallic element that has a relative atomic mass of 85.47. There are two forms of naturally occurring rubidium isotopes, ^{85}Rb and ^{87}Rb .

^{87}Rb is slightly radioactive.

What percentage of naturally occurring rubidium is radioactive?

- A** 11.8 **B** 23.5 **C** 35.3 **D** 76.5

- 2 10 cm³ of hydrocarbon was burnt with 100 cm³ of oxygen gas which is in excess. The resulting mixture was cooled to room temperature and the residual gases occupied a volume of 80 cm³. After passing through sodium hydroxide, the volume of gas remaining is 40 cm³.

What is the formula of the hydrocarbon?

- A** C₃H₆ **B** C₃H₈ **C** C₄H₈ **D** C₄H₁₀

- 3 Which of the following is true when equimolar amounts of NH₃ and BF₃ is reacted?

- 1 a dative covalent bond is formed between NH₃ and BF₃
- 2 N in NH₃ accepts a lone pair of electrons from B in BF₃
- 3 the shape around N in NH₃ changes from trigonal planar to tetrahedral
- 4 after mixing, the bond angle around B in BF₃ decreased from 120° to 109°

- A** 1 and 2 only
B 1 and 4 only
C 1, 3 and 4 only
D 3 and 4 only

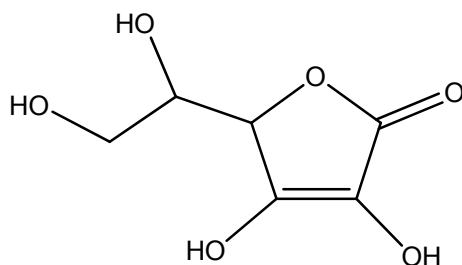
- 4 *Use of Data Booklet is relevant to this question.*

How many hydrogen atoms are present in 111 g of propanoic acid?

[*L* = Avogadro constant]

- A** *L* **B** 1.5*L* **C** 7.5*L* **D** 9*L*

- 5 Ascorbic acid, commonly known as vitamin C, is an important dietary supplement as it is essential in the repairing of tissues and can also function as an antioxidant.



vitamin C

Which of the following shows the correct number of sigma and pi bonds present in one molecule of vitamin C?

	sigma bonds	pi bonds
A	11	2
B	16	2
C	20	1
D	20	2

- 6 The table below shows the boiling points of some halogenoalkanes.

halogenoalkane	boiling point/ $^{\circ}\text{C}$
$\text{CH}_3\text{CH}_2\text{Cl}$	12.3
$\text{CH}_3\text{CH}_2\text{Br}$	34.8
$\text{CH}_3\text{CH}_2\text{I}$	70.0

Which of the following statements correctly explains the difference in the boiling point?

- A** the bond energy of C-X bond decreases from C-Cl to C-I
- B** the strength of instantaneous dipole-induced dipole attraction increases from $\text{CH}_3\text{CH}_2\text{Cl}$ to $\text{CH}_3\text{CH}_2\text{I}$
- C** the strength of permanent dipole-permanent dipole attraction increases from C-Cl to C-I
- D** the electronegativity difference between the halogen and carbon increases from C-Cl to C-I

- 7 The following information is true for substance **Z**.
- It exists as a solid state at room temperature and pressure
 - It has a regular lattice structure at room temperature and pressure
 - It does not conduct electricity in both solid and molten state

Which of the following could be substance **Z**?

- A** iodine
B graphene
C aluminium
D sodium chloride

- 8 *Use of the Data Booklet is relevant to this question.*

Radioactive carbon-14, ^{14}C , is often used to date archaeological and geological samples. Which species has the same number of neutrons and the same number of electrons as an atom of ^{14}C ?

- A** $^{18}\text{Ne}^{2+}$ **B** $^{17}\text{F}^{+}$ **C** $^{16}\text{O}^{2+}$ **D** $^{14}\text{N}^{+}$

- 9 *Use of the Data Booklet is relevant to this question.*

The successive ionisation energies, in kJ mol^{-1} , of an element **X** are given below.

940	2045	3060	4140	6590	7895	14990
-----	------	------	------	------	------	-------

What is **X**?

- A** ^{33}As **B** ^{34}Se **C** ^{39}Y **D** ^{85}At

- 10 Element **Y** is in period 3 and has melting point higher than the element before it. **Y** forms a chloride with low melting point which dissolves in water giving a solution of pH 2. What is **Y**?

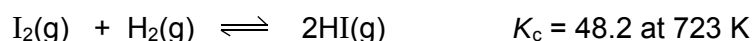
- A** magnesium **B** aluminium **C** silicon **D** phosphorus

- 14 Use of the Data Booklet is relevant to this question.

Methanol (M_r 32) is a preferred fuel for high performance racing cars. When 1.00 g of methanol was burnt to heat up a copper can with 200 g of water, it was found that the temperature of the water rose by 27 °C.

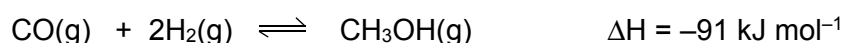
Assuming there is no heat lost to the copper can or surrounding, which value for the enthalpy change of combustion of methanol is given by these results?

- A -8026 kJ mol⁻¹
 B -722 kJ mol⁻¹
 C -251 kJ mol⁻¹
 D -113 kJ mol⁻¹
- 15 Equal amounts of hydrogen and iodine were kept at a constant temperature of 723 K until equilibrium was reached. After which, rapid cooling was conducted to extract the iodine present for titration, and it was found that 0.07 mol of iodine was present.



What is the amount in moles of HI at equilibrium?

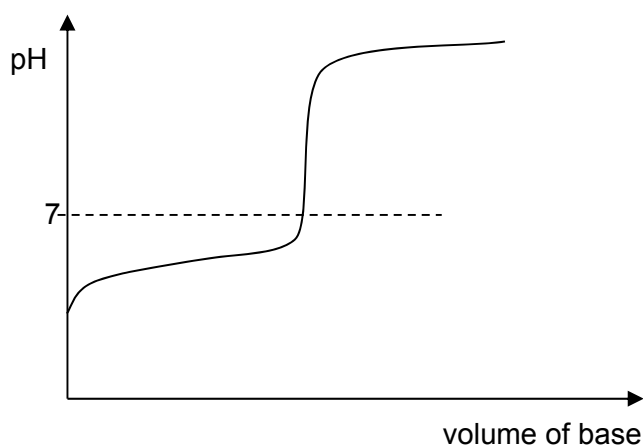
- A 0.236 B 0.243 C 0.486 D 1.84
- 16 Methanol can be manufactured industrially, using a copper oxide catalyst at temperature of 575 K, according to the following reaction.



Which of the following statements are correct?

- 1 The catalyst increases the equilibrium constant.
 - 2 The catalyst increases the equilibrium yield.
 - 3 At lower temperature the equilibrium yield will increase.
 - 4 At lower temperature the equilibrium constant will increase.
- A 1 and 2 only
 B 2 and 3 only
 C 1 and 4 only
 D 3 and 4 only

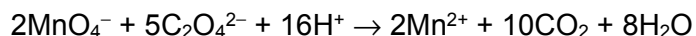
- 17 The Haber process is a key reaction in the industry for producing ammonia. Which statement about Haber process is **not** correct?
- A Iron is used as a heterogeneous catalyst to increase rate of production.
 B Temperature of around 500 °C is used to increase the yield of ammonia.
 C High pressure is preferred for this reaction as it increases the yield of ammonia.
 D High pressure and temperature leads to an increase in the rate of production.
- 18 What is the final pH of the solution when equal volumes of 0.030 mol dm⁻³ of Ba(OH)₂(aq) is mixed with 0.030 mol dm⁻³ of HCl(aq)?
- A 1.5 B 1.8 C 12.2 D 12.5
- 19 Visual indicators are often used to determine the end point of acid-base titrations. A titration was carried out between a weak acid and a strong base and the titration graph obtained is shown below:



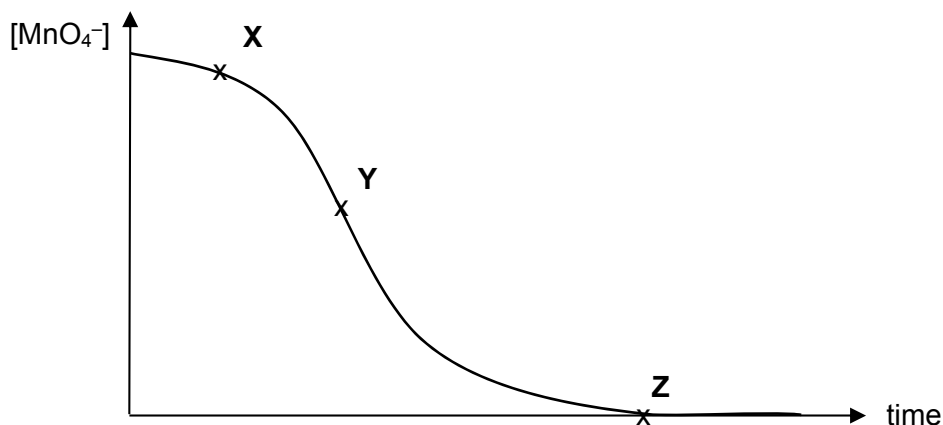
Which of the following is a suitable indicator for the above titration?

	Indicator	pH at which colour changes
A	Methyl violet	0 - 1
B	Bromocresol green	3 - 6
C	Bromothymol blue	6 - 7
D	Thymolphthalein	9 - 11

- 20** The reaction of manganate(VII), MnO_4^- , with oxalate, $\text{C}_2\text{O}_4^{2-}$ is a type of autocatalytic reaction whereby the product, manganate(II) ions, is the catalyst for the reaction.



The following graph was obtained when $[\text{MnO}_4^-]$ was plotted against time.



Which of the following can be deduced from the graph?

- A** Rate at point **X** is very slow indicating that the temperature is low.
 - B** Rate at point **Y** fast due to the production of the catalyst.
 - C** Rate at point **Z** is slow as the catalyst is saturated.
 - D** Rate at point **Z** is slow as the reaction is zero order with respect to $[\text{MnO}_4^-]$.
- 21** When aqueous solutions of HCO_2H and Br_2 are reacted together, initial rate of reaction was found to be $0.080 \text{ mol dm}^{-3} \text{ s}^{-1}$.

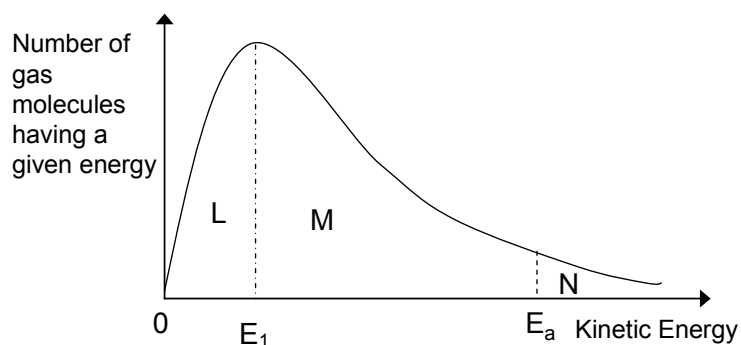
Given that the rate equation is:

$$\text{rate} = k[\text{Br}_2][\text{HCO}_2\text{H}]$$

Which of the following shows the initial rate of reaction when the original reaction mixture was mixed with an equal volume of water?

- A** $0.010 \text{ mol dm}^{-3} \text{ s}^{-1}$
- B** $0.020 \text{ mol dm}^{-3} \text{ s}^{-1}$
- C** $0.040 \text{ mol dm}^{-3} \text{ s}^{-1}$
- D** $0.080 \text{ mol dm}^{-3} \text{ s}^{-1}$

- 22** The Boltzmann distribution at a constant temperature is shown below:
 E_1 and E_a are fixed energy values.



If the temperature is decreased by 10°C , what happens to the size of the areas labelled **L**, **M** and **N**?

	L	M	N
A	decreases	decreases	decreases
B	decreases	increases	decreases
C	increases	decreases	decreases
D	increases	decreases	increases

- 23** The following table gives the approximate dimensions of certain substances.

	substance	dimensions
1	Fullerene (buckyball)	Diameter 0.71 nm
2	Red blood cell	Diameter of $8\ \mu\text{m}$, thickness of $2.5\ \mu\text{m}$
3	DNA strand	Diameter 2 nm, length up to 5 cm
4	Adenovirus	Diameter 90 nm

($1\ \mu\text{m} = 1000\ \text{nm}$)

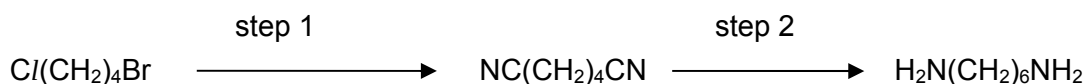
Which of these would be considered nanoparticles?

- A** 1 and 2 only
B 1, 2 and 4 only
C 1 and 4 only
D 2 and 3 only

- 24** Dishwasher pouches are small convenient packets that contain the detergent needed to be used in dishwashers. Such pouches are sturdy enough to be easily transported and safely handled by hand, yet dissolve easily when it is time to wash dishes. What could be the polymer that is used for the pouches?

- A** Polyvinyl alcohol
- B** Polyvinyl chloride
- C** Poly(diallyl phthalate)
- D** Poly(ethylene terephthalate)

- 25** The reaction scheme outlines the production of one of the monomers of nylon 66 from $\text{Cl}(\text{CH}_2)_4\text{Br}$.



What are the types of reactions for each stage?

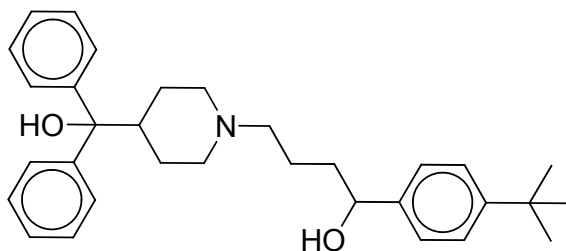
	step 1	step 2
A	substitution	oxidation
B	substitution	reduction
C	addition	addition
D	addition	reduction

- 26** Which of the following are true regarding polyester fabrics in comparison to polyamide fabrics?

- 1 Polyester fabrics tend to crease less easily than polyamide fabrics.
- 2 Polyester fabrics tend to be more easily stained by oil than polyamide fabrics.
- 3 Polyester fabrics tend to absorb more sweat than polyamide fabrics.

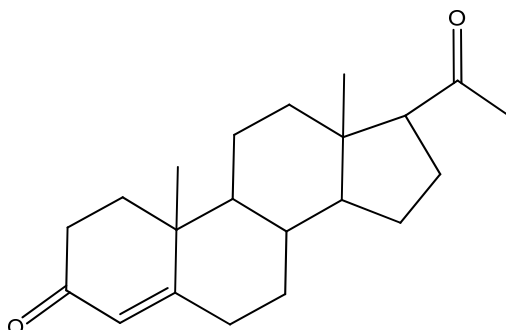
- A** 1 and 2 only
- B** 2 and 3 only
- C** 3 only
- D** 1, 2 and 3 only

- 27 *Terfenadine* is an antihistamine formerly used for the treatment of allergic conditions. The structure of *terfenadine* is shown below:



Which functional group is not present in *terfenadine*?

- A amine
 - B tertiary alcohol
 - C secondary alcohol
 - D phenol
- 28 Progesterone is a hormone which is involved in menstrual cycle and pregnancy. The following structure shows one molecule of progesterone.



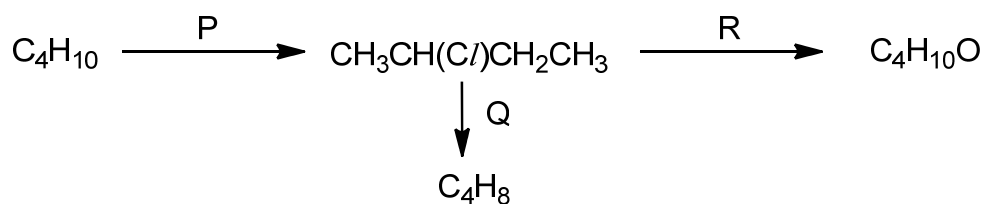
What is the molecular formula for the progesterone hormone?

- A $C_{19}H_{14}O_2$
- B $C_{19}H_{22}O_2$
- C $C_{20}H_{30}O_2$
- D $C_{21}H_{30}O_2$

29 Which of the following alcohols can be formed when a ketone is reduced with NaBH_4 dissolved in methanol?

- A 2-methylpropan-1-ol
- B 2-methylbutan-2-ol
- C 3-methylbutan-2-ol
- D 2,3-dimethylbutan-2-ol

30 The diagram shows a reaction scheme involving 2-chlorobutane, $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{CH}_3$.



Which of the following is true regarding the reaction scheme above?

- 1 both reactions P and R are substitution reactions
- 2 reaction Q is a reduction reaction
- 3 NaOH can be used in both reactions Q and R
- 4 two constitutional isomers with the molecular formula of C_4H_8 can be formed from reaction Q

- A 1 and 2 only
- B 1 and 3 only
- C 1, 3 and 4 only
- D 2, 3 and 4 only